

Amel Derras-Chouk

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Education

The Graduate Center, City University of New York New York, NY
PhD, Physics Sept. 2022
Dissertation: *Stability of Two-Dimensional Magnetic Skyrmions*
Advisors: Dist. Prof. Eugene M. Chudnovsky and Prof. Dmitry A. Garanin

Tufts University, School of Engineering Medford, MA
Bachelor of Science: Double major in Mechanical Engineering and Physics May 2015

Research Experience

Columbia Climate School New York, NY
Postdoctoral Research Fellow Sept. 2024 – present

- Develop physical models to explain emergent ordered behavior in tropical anvil clouds
- Process remote sensing data to validate analytical models with cloud observations

The City College of New York New York, NY
Postdoctoral Researcher Sept. 2022 – Sept. 2024

- Combine remote sensing and physical models to quantify cumulus convection, estimating uncertain atmospheric variables like vertical velocity and mass flux

CUNY Graduate Center Bronx, NY
Graduate Researcher Jun. 2017 – Aug. 2022

- Developed classical and quantum theories to model two-dimensional spin systems
- Parallelized thousands of computations using multithreading and distributed computing
- Analyzed program outputs by visualizing results and performing statistical tests
- Presented results to audiences of scientific backgrounds at national conferences

NASA Climate Change Research Initiative New York, NY
Graduate Student Research Assistant Oct. 2021 – Aug. 2022

- Automated retrieval and analysis of over 40 years of Landsat optical data using Python
- Proposed a methodology based on literature to measure flooding using Sentinel-1 data
- Integrated data from NOAA, USGS, ASF, and citizen scientists into a cohesive workflow
- Tailored presentations to heterogeneous audiences of experts and local teachers

Manuscripts in Preparation

- Derras-Chouk, A., Luo, Z. J., and Elsaesser, G.. A Tropical Climatology of Precipitation Efficiency in Deep Convective Clouds.
- Derras-Chouk, A, Elsaesser, G, Luo, Z.J., Matsui, T., Prein, A., and Wu, J. Estimating Updraft Vertical Velocity from Temperature, Pressure, and Latent Heating.

Publications

- Derras-Chouk, A and Z.J. Luo, 2024: A geostationary satellite-based approach to estimate convective mass flux and revisit the hot tower hypothesis. *Surveys in Geophysics*, 45(6), 1959-1977.
- Braneon, C., L. Ortiz, D. Bader, N. Devineni, P. Orton, B. Rosenzweig, T. McPhearson, L. Smalls-Mantey, V. Gornitz, T. Mayo, S. Kadam, H. Sheerazi, E. Glenn, L. Yoon, A. Derras-Chouk, J. Towers, R. Leichenko, D. Balk, P. Marcotullio and R. Horton, 2024: NPCC4: New York City climate risk information 2022—observations and projections, *Annals of the New York Academy of Sciences*.
- Derras-Chouk, A., E. M. Chudnovsky, and D. A. Garanin, 2022: Dynamics of the collapse of a ferromagnetic skyrmion in a centrosymmetric lattice. *Physical Review B*, **105**, 134432.
- Derras-Chouk, A., E. M. Chudnovsky, and D. A. Garanin, 2021: Quantum States of a Skyrmion in a Two-Dimensional Antiferromagnet. *Physical Review B*, **103**, 224423-(8).
- Derras-Chouk, A., and E. M. Chudnovsky, 2021: Skyrmions Near Defects. *Journal of Physics: Condensed Matter*, **33**, 195802-(10).
- Derras-Chouk, A., E. M. Chudnovsky, and D. A. Garanin, 2019: Thermal Collapse of a Magnetic Skyrmion. *Journal of Applied Physics*, **126**, 083901. *(Featured Selection)*
- Derras-Chouk, A., E. M. Chudnovsky, and D. A. Garanin, 2018: Quantum Collapse of a Magnetic Skyrmion. *Phys. Rev. B*, **98**, 024423. *(Editor's Suggestion)*
- Derras-Chouk, A., E. M. Chudnovsky, D. A. Garanin, and R. Jaafar, 2018: Graphene Cantilever under Casimir Force. *Journal of Physics D*, **51**, 19.

Selected Presentations

- Derras-Chouk, A., Z. J. Luo, and G. Elsaesser, 2025: Estimating convective mass flux, detrainment, and precipitation efficiency using CloudSat, Geostationary, and IMERG data. Oral presentation delivered at CloudSat Calipso Science Team Meeting, Fort Collins, CO, USA
- Derras-Chouk, A., and Z. J. Luo, 2023: A Satellite-Based Approach to Estimating Convective Mass Flux and Revisiting the Hot Tower Hypothesis. *AGU*, San Francisco, CA.
- Derras-Chouk, A., E. M. Chudnovsky, and D. A. Garanin, 2019: Quantum Collapse of a Magnetic Skyrmion. *APS March Meeting*, Boston, MA.
- Derras-Chouk, A., E. M. Chudnovsky, D. A. Garanin, and R. Jaafar, 2018: Graphene Cantilever under Casimir Force. *APS March Meeting*, Los Angeles, CA.

Teaching Experience

Lehman College

The Bronx, NY

Adjunct Lecturer

Aug. 2017 – May 2021

- Taught physics labs to over 50 students a semester, emphasizing experimental design
- Assessed student understanding through written assignments and problem-solving sessions
- Redesigned seven experiments for remote learning and incorporated student feedback

Success Academy Charter Schools

The Bronx, NY

Computer Science Lead Teacher

Aug. 2015 – May 2016

- Engaged students in a progressive pedagogical approach to computer science
- Developed both professional and teaching skills in monthly network-wide trainings

Tufts Office of Physics and Astronomy
Teaching Assistant

Medford, MA
Sept. 2014 – May 2015

- Tested and set up electronic equipment for undergraduates in biweekly lab sessions
- Enforced lab safety standards and replaced faulty, unsafe components when necessary

Community Service

Word Up Community Bookshop
Collective Member & Donation Analyst

New York, NY
Sept. 2017 – Jan. 2024

- Streamline collection of donation data from multiple sources by automating with Python
- Summarize recent donation activity to present to other members of the collective

Honors and Awards

Graduate Center Dissertation Fellowship
Physics Tithe Fellowship

2020-2021
2018-2022

Skills

Programming languages: Julia, Python, C++, Java

Software: Mathematica, Matlab, LabView, SolidWorks, AutoCAD